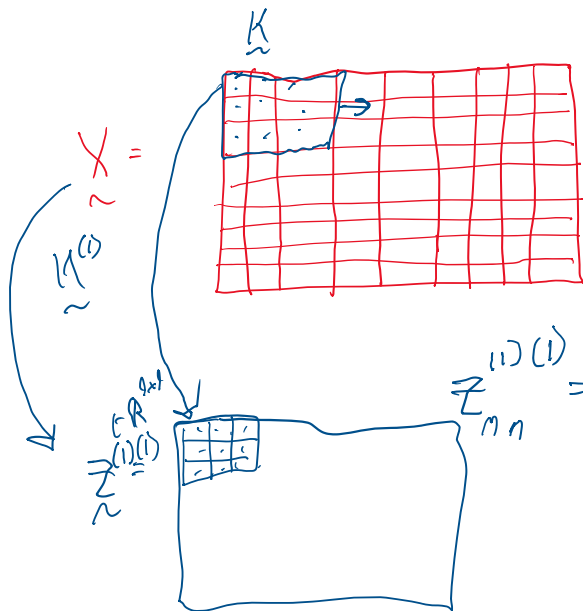


- Suppose we have a kernel/filter $\tilde{K} \in \mathbb{R}^{3 \times 3}$ + image $\tilde{X} \in \mathbb{R}^{9 \times 9}$

$$\tilde{K} \text{ has a } 3 \times 3 \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}$$

Structure where $\tilde{K}$ has the weights for this layer of the NN



$$Z_{m,n}^{(1)(1)} = \sum_i \sum_j K_{ij}^{(1)(1)} X_{m-i, n-j} + b^{(1)(1)}$$

- When we apply CNNs, we typically apply multiple filters/kernels to the image

• Suppose we have $\tilde{X} \in \mathbb{R}^{9 \times 9}$, + we apply 3 kernels/filters that are all 3×3 , i.e. $\tilde{K}^{(1)(2)} \in \mathbb{R}^{3 \times 3}$, $2=1,2,3$ (so each filter has 9 weights + 1 bias)

→ first layer: $(9+1) \times 3 = 30$ parameters

$$\Rightarrow Z_{m,n}^{(1)(2)} = \sum_i \sum_j K_{ij}^{(1)(2)} \tilde{X}_{m-i, n-j} + b^{(1)(2)}$$

$$A^{(1)(2)} = \sigma^{(1)}(\tilde{Z}^{(1)(2)})$$

⇒ we get 3 new 9x9 images, $\tilde{A}^{(1)(1)}$, $\tilde{A}^{(1)(2)}$, $\tilde{A}^{(1)(3)}$

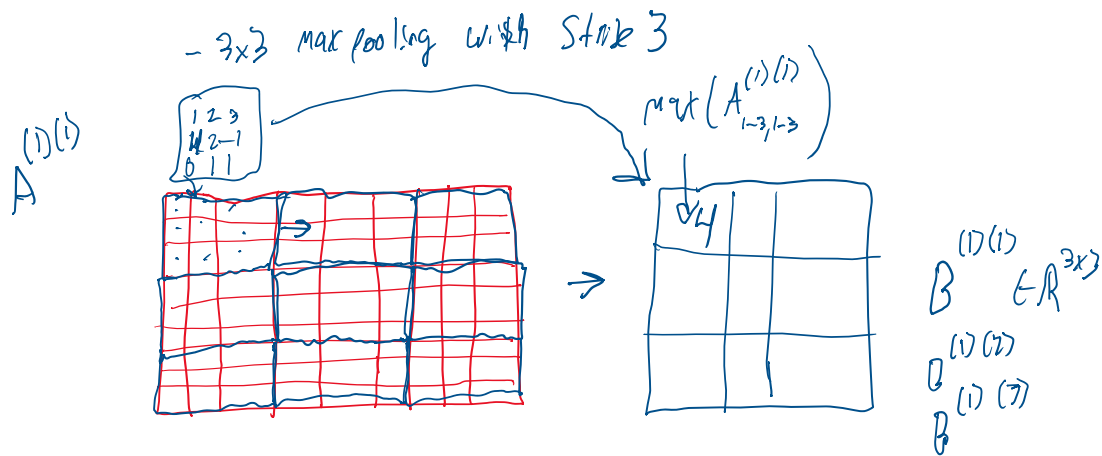
$\Rightarrow$ we get 3 new 9×9 images, $\tilde{A}^{(1)(1)}$, $\tilde{A}^{(1)(2)}$, $\tilde{A}^{(1)(3)}$

Pooling

- Pooling takes our outputs from the convolutional layer & performs an operation to reduce the size of the output.

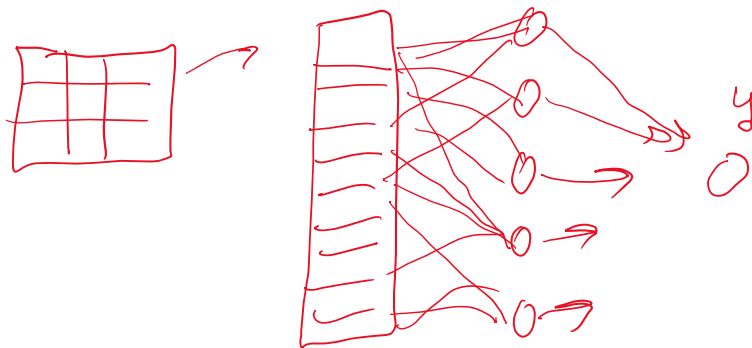
• We select the pooling size (e.g., 2×2 or 3×3)

◦ Then the pooling applies an operation, e.g. the max, to that image over the "pool" size



- After applying convolutions & pooling, we get multiple versions of a smaller image or input data

- At the last layer, we typically "concatenate" or "flatten" the output



→ standard CNN approach

